

Miguel Couto de Almeida

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Personal Website | Linkedin | Github

Summary

Full-Stack Software Engineer with experience in R&D developing distributed, microservices-based systems. Skilled in event-driven architectures, with practical experience applying cryptographic concepts in authentication systems.

Education

Instituto Superior de Engenharia de Lisboa, BSc in Computer Science and Engineering Sept 2019 – Jul 2023

Experience

Staff Consultant / Full-Stack Software Engineer, EY – Lisbon, Portugal May 2025 – Present

Contributed to the development and implementation of enterprise software solutions for financial institutions, focusing on KYC/AML platforms, digital transformation, and distributed systems.

- Implemented EY's in-house KYC/AML platform for an overseas banking client, contributing to backend services, system integration, and production deployment.
- Participated in a consulting engagement advising one of Portugal's largest banks on the adoption of digital signature solutions.
- Developed and maintained EY's internal KYC/AML platform, building REST APIs and event-driven microservices using Java, Spring Boot, and Kafka.
- Collaborated with cross-functional teams to deliver scalable, cloud-native applications in an Agile environment.

Technologies and Tools: Java 21/25, Spring Boot, REST APIs, Kafka, React, TypeScript, MongoDB, Elasticsearch, MySQL, Kubernetes, Lens, Azure DevOps, Docker

Software Development Trainee, Infinera (R&D Department) – Lisbon, Portugal Sept 2023 – Jul 2024

Contributed to a large-scale, microservices-based platform for planning, real-time control, and monitoring of multi-layer communication networks used by telecom operators.

- Implemented testing strategies (unit, integration, end-to-end), improving reliability across distributed services.
- Developed and maintained backend services using Java, Spring Boot, gRPC, and Kafka in a production-grade distributed system.
- Diagnosed and resolved software defects, improving system stability and performance in a high-availability environment.
- Authored technical documentation to support onboarding and cross-team collaboration.

Technologies and Tools: Java 17, Spring Boot, gRPC, Kafka, Docker, OracleDB, JPA, Hibernate

Co-Founder & Host, Aceita Bitcoin Jan 2020 – Present

Co-founded Aceita Bitcoin, an initiative focused on advancing Bitcoin education and adoption in Portugal through content, community, and technical experimentation, reaching over 2,000 users.

Exploring the platform as a foundation for a “network state” model built on a Bitcoin standard, integrating technology, governance, and digital community building.

Recognised by the Portuguese Mises Institute and featured in Região de Leiria journal for contributions to discussions on Bitcoin economics.

Technologies and Tools: Svelte, Rollup, Javascript, NPM

Projects

Battleships App - Full-Stack Web and Mobile Application (Android) github.com

Developed an online, cross-platform version of the classic Battleships game, allowing players to connect and

compete.

- Technologies and Tools: Kotlin, Spring Boot, React, Typescript, Android Development, Jetpack Compose, Docker, Nginx, PostgreSQL

Landmark Identification App - Distributed System for Cloud Computing Tasks

github.com

Designed and implemented a system to detect and verify famous landmarks in images, adapting to load variations using Google Cloud Platform services.

- Technologies and Tools: Java, Cloud Functions, Pub/Sub, GCP, Compute Engine, Vision API, gRPC, Protocol Buffers, Google Maps API, Cloud Firestore

JSON Parser Library - Custom JSON Parsing Library in Kotlin

github.com

Developed a JSON parser library using JavaPoet for code generation and JMH for benchmarking. Leveraged Kotlin's reflection library to enhance parsing efficiency and flexibility across diverse data structures.

- Technologies and Tools: Kotlin, JavaPoet, JMH, Kotlin Reflection

Technologies

Languages: Java, Kotlin, JavaScript, Typescript, Python, SQL, PL/pgSQL

Backend & Systems: Spring Boot, NodeJS, gRPC, Kafka, REST API, OpenAPI, Protobuf, Siren Hypermedia

Frontend: React, Svelte, JetPack Compose (Android & Desktop), Bootstrap

Cloud & DevOps: Google Cloud Platform, Docker, Kubernetes, Azure DevOps, Nginx

Databases: PostgreSQL, MySQL, MongoDB, Elasticsearch

Speaking Engagements

Speaker, Bitcoin Atlantis Conference – Funchal, Madeira

Mar 2024 – Mar 2024

As the Co-Founder of our Bitcoin Education Project, I had the unique opportunity to take the lead in planning and arranging an entire day of educational content for the Portuguese audience at the Bitcoin Atlantis Conference in Funchal, Madeira, in March 2024. With more than 3,000 attendees, invited by André Loja, the founder of FREE Madeira, I organised a comprehensive series of lectures and workshops designed to deepen understanding and practical use of Bitcoin, such as:

- Defining the Problem and Designing the Solution with Bitcoin
- How Bitcoin Works and The Bitcoin Standard
- On-Chain and Lightning Network Usage for Merchants and End Users
- Self-Custody, Security, and Privacy Concerns
- Bitcoin Inheritance Solutions

Speaker, Futuro Bitcoin – Lisbon/Porto, Portugal

May 2025

Delivered talks on *Social and Technical Scalability in Bitcoin* and the intersection of Software Systems and Economic Design, to an audience of 75+ attendees.

- Presented alongside Prof. Dr. Renato Amoedo Nadier Rodrigues and Prof. Dr. Guilherme Villela de Viana Bandeira in a multidisciplinary setting combining engineering, law, and economics.
- Contributed to thematic sessions covering Bitcoin layers, mining and energy production, and long-term technological development.
- Participated in workshops on self-custody and Bitcoin inheritance, helping bridge technical concepts with practical user adoption.